LUNA ZEA-REDONDO, Ph.D.

Computational Systems Biologist

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SUMMARY

Computational Systems Biologist focused on **quantitative characterization** and analysis of **complex biological systems**. PhD in Molecular Systems Biology with **10+ years of research experience** and deep expertise in **time-series analysis**, regulatory network inference, and multi-omics data integration. I excel at designing **end-to-end** projects that translate biological questions into clear computational strategies in cross-functional environments. Eager to move toward more **model-centric frameworks** for understanding complex biological systems.

PROFESSIONAL EXPERIENCE

Senior Research Scientist | Computational Systems Biology **Oct 2025 – present**
Berlin Institute for Medical Systems Biology (MDC-BIMSB) *Berlin, Germany*

- **Led** the end-to-end delivery of a **first-author multi-omics study** to final submission, integrating project design, experimental execution, computational analysis and scientific narrative in alignment with senior stakeholders.
- **Initiated** the design and execution of **two collaborative research projects** derived from my PhD work, coordinating delivery across **four international cross-functional teams**.

Ph.D. Researcher | Computational Systems Biology **Jan 2019 – Oct 2025**
Berlin Institute for Medical Systems Biology (MDC-BIMSB) *Berlin, Germany*

- **Designed and executed** a **single-cell multi-omics** study (18 samples, >150,000 cells, 6 time points) covering mapping, QC, differential analyses, time-course modeling, and regulatory network inference to uncover mechanisms of cellular memory in **neuronal systems**.
- **Generated and analyzed** over **10 large-scale genomic datasets** contributed to **five scientific publications**.
- **Co-led** a collaborative project resulting in a **co-first-author *Nature*** (2021) publication (> **200 citations**), integrating scRNA-seq, scATAC-seq, and 3D genome architecture across brain cell types.
- **Developed** robust, reusable **analytical workflows** adopted across the group, improving reproducibility and standardizing analysis practices.
- **Awarded a competitive** DFG-funded **PhD scholarship,** a collaborative venture between Humboldt and Duke Universities that recognized academic excellence and research capabilities.
- **Awarded an outstanding research contributions prize** and a competitive **travel fellowship**; delivered **two selected talks** and presented **five scientific posters** at international conferences.
- **Mentored two researchers** in computational analysis strategies and served as **co-reviewer for five journals**.

Junior Scientist | Cancer Genomics & Computational Biology **Feb 2017 – Aug 2018**
New York University Medical School (NYUMS) *New York, US*

- **Conducted** a project investigating MED12 mutations in chronic lymphocytic leukemia, generating and analyzing **+20 ChIP-, RNA-, and ATAC-seq datasets** from primary patient blood samples and cell lines.
- **Performed** computational analyses of over **10 ChIP-seq and ATAC-seq datasets** for studies on inflammation and leukemic transformation contributing to **two high impact publications** (*Nature Immunology* and *Cell Stem Cell*).

Research Fellow | Molecular Biology & Functional Genomics **Oct 2015– Sept 2016**
European Molecular Biology Laboratory (EMBL) *Rome, Italy*
Max-Planck-Institute for Molecular Genetics (MPIMG) *Berlin, Germany*

- Conducted **hands-on molecular biology** work bridging wet-lab and computational expertise across two leading European research institutes.

EDUCATION

PhD in Systems Biology • Humboldt University of Berlin • Berlin, Germany (*Summa cum laude*) • **2025**
MSc in Bioinformatics • Autonomous University of Barcelona • Barcelona, Spain • **2017**
BSc in Biochemistry and Molecular Biology • University of Córdoba • Córdoba, Spain • **2015**

SKILLS

Programming & Development: Python, R, Shell, Git, Docker.
Statistical Analysis & Machine Learning: Statistical modeling, time-series analysis, dimensionality reduction, unsupervised learning (clustering, PCA).
Core Data Stack: NumPy, Pandas, Scikit-learn.
Bioinformatics: Multi-omics analysis, single-cell analysis (Seurat, Scanpy), genomics pipelines (STAR, Bowtie2, MACS2, deepTools), regulatory network inference.
Project & Communication: Cross-functional teamwork, project planning, mentoring, scientific writing, peer review, conference presentation.
Experimental Background: Hands-on experience with genomic library prep and molecular biology techniques, ensuring deep understanding of data generation and quality requirements.
Languages: English (fluent), Spanish (native), German (intermediate).

📊 Full project details, publications, and additional materials at: **red-moonx.github.io**